

CS182 HW1

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June 2025

Problem 1

(a)

We look at the recurrence relation:

$$w_{t+1} = (1 - 2\eta\sigma^2)w_t + 2\eta\sigma y$$

Let $r = 1 - 2\eta\sigma^2$. For convergence, we need the homogeneous part to decay:

$$|r| < 1 \Rightarrow -1 < 1 - 2\eta\sigma^2 < 1 \Rightarrow 0 < \eta < \frac{1}{\sigma^2}$$

The recurrence is stable if and only if $\eta < \frac{1}{\sigma^2}$.

(b)

Let $w^* = \frac{y}{\sigma}$, the optimal solution. The recurrence:

$$w_{t+1} - w^* = r(w_t - w^*) \Rightarrow |w_t - w^*| = |r|^t |w_0 - w^*|$$

To get within a factor $(1 - \epsilon)$ of w^* :

$$|r|^t \leq \epsilon \Rightarrow t \geq \frac{\log \epsilon}{\log |r|} = \frac{\log \epsilon}{\log |1 - 2\eta\sigma^2|}$$

(c)

We now solve:

$$\begin{bmatrix} \sigma_\ell & 0 \\ 0 & \sigma_s \end{bmatrix} \begin{bmatrix} w[1] \\ w[2] \end{bmatrix} = \begin{bmatrix} y[1] \\ y[2] \end{bmatrix}$$

Gradient descent update:

$$w_t = w_{t-1} + \eta \cdot \nabla L(w_{t-1}) \Rightarrow w_{t+1}^{[i]} = (1 - 2\eta\sigma_i^2)w_t^{[i]} + 2\eta\sigma_i y^{[i]}$$

Using the previous result, each coordinate converges if:

$$\eta < \frac{1}{\sigma_i^2}$$

We converge if $\eta < \frac{1}{\sigma_\ell^2}$, where $\sigma_\ell = \max(\sigma_\ell, \sigma_s)$. The larger σ limits the learning rate.

(d)

Let $r_i = 1 - 2\eta\sigma_i^2$. The smaller $|r_i|$ is, the faster convergence is.

$$\sigma_\ell \gg \sigma_s \Rightarrow |1 - 2\eta\sigma_\ell^2| \gg |1 - 2\eta\sigma_s^2|$$

So: - $w^{[2]}$ (associated with σ_s) converges slowly. - $w^{[1]}$ (with σ_ℓ) converges faster.

(e)

The overall speed is bottlenecked by the slower direction:

$$\eta < \frac{1}{\sigma_\ell^2} \Rightarrow \text{Use } \eta = \frac{1}{\sigma_\ell^2} \text{ (just under)}$$

(f)

If intermediate σ_i exist, learning rate is still limited by the largest σ_i , the number of slower dimensions increases, it reduces the benefit of fast directions, as most variables converge slowly, and convergence becomes slower overall.

(g)

In general least squares,

$$\min_w \|Xw - y\|^2 \Rightarrow \text{gradient } \nabla = -2X^\top(y - Xw)$$

Using SVD, $X = U\Sigma V^\top$, so problem becomes:

$$\min_z \|\Sigma z - U^\top y\|^2, \quad \text{where } z = V^\top w$$

Each dimension in z corresponds to a scalar problem like above. The singular values σ_i act like σ in the scalar case.

Problem 2

(a)

Let us expand the objective:

$$\|W_1(A\mathbf{x} - \mathbf{b})\|_2^2 = (A\mathbf{x} - \mathbf{b})^T W_1^T W_1 (A\mathbf{x} - \mathbf{b})$$

$$\|W_2(\mathbf{x} - \mathbf{c})\|_2^2 = (\mathbf{x} - \mathbf{c})^T W_2^T W_2 (\mathbf{x} - \mathbf{c})$$

So the total cost function becomes:

$$J(\mathbf{x}) = (A\mathbf{x} - \mathbf{b})^T W_1^T W_1 (A\mathbf{x} - \mathbf{b}) + (\mathbf{x} - \mathbf{c})^T W_2^T W_2 (\mathbf{x} - \mathbf{c})$$

Taking the gradient with respect to $\mathbf{x}$,

$$\nabla_{\mathbf{x}} J = 2A^T W_1^T W_1 (A\mathbf{x} - \mathbf{b}) + 2W_2^T W_2 (\mathbf{x} - \mathbf{c})$$

Setting the gradient to zero:

$$A^T W_1^T W_1 A \mathbf{x} - A^T W_1^T W_1 \mathbf{b} + W_2^T W_2 \mathbf{x} - W_2^T W_2 \mathbf{c} = 0$$

$$(A^T W_1^T W_1 A + W_2^T W_2) \mathbf{x} = A^T W_1^T W_1 \mathbf{b} + W_2^T W_2 \mathbf{c}$$

Therefore, the optimal solution is

$$\mathbf{x}^* = (A^T W_1^T W_1 A + W_2^T W_2)^{-1} (A^T W_1^T W_1 \mathbf{b} + W_2^T W_2 \mathbf{c})$$

(b)

We want to write the problem as:

$$\min_{\mathbf{x}} \|C\mathbf{x} - \mathbf{d}\|_2^2$$

Let:

$$C = \begin{bmatrix} W_1 A \\ W_2 \end{bmatrix}, \quad \mathbf{d} = \begin{bmatrix} W_1 \mathbf{b} \\ W_2 \mathbf{c} \end{bmatrix}$$

Then:

$$\|C\mathbf{x} - \mathbf{d}\|_2^2 = \|W_1(A\mathbf{x} - \mathbf{b})\|_2^2 + \|W_2(\mathbf{x} - \mathbf{c})\|_2^2$$

So this representation matches the original problem.

Using the OLS solution:

$$\mathbf{x}^* = (C^T C)^{-1} C^T \mathbf{d}$$

$$= (A^T W_1^T W_1 A + W_2^T W_2)^{-1} (A^T W_1^T W_1 \mathbf{b} + W_2^T W_2 \mathbf{c})$$

which matches the result from part (a).

(c)

We want to reduce the general problem to the standard ridge regression solution:

$$\mathbf{x}^* = (A^T A + \lambda I)^{-1} A^T \mathbf{b}$$

To do this, choose:

$$W_1 = I, \quad W_2 = \sqrt{\lambda} I, \quad \mathbf{c} = \mathbf{0}$$

Then the original problem becomes:

$$\|A\mathbf{x} - \mathbf{b}\|_2^2 + \lambda \|\mathbf{x}\|_2^2$$

This just the standard ridge regression objective. Plugging into the general solution:

$$\mathbf{x}^* = (A^T A + \lambda I)^{-1} A^T \mathbf{b}$$

Problem 3

We must show that the two forms of the Ridge Regression solution

$$\hat{w} = (X^T X + \lambda I)^{-1} X^T y \quad (5)$$

and

$$\hat{w} = X^T (X X^T + \lambda I)^{-1} y \quad (6)$$

are identical by interpreting Ridge Regression as a MAP estimation problem where both W and Y are jointly Gaussian.

Assume the prior on weights is:

$$W \sim \mathcal{N}(0, I)$$

Assume the likelihood model:

$$Y = XW + \sqrt{\lambda}N, \quad N \sim \mathcal{N}(0, I) \Rightarrow Y | W \sim \mathcal{N}(XW, \lambda I)$$

From the above, since both W and N are Gaussian and independent, the joint distribution $[W, Y]$ is jointly Gaussian:

$$\begin{bmatrix} W \\ Y \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} I & \Sigma_{WY} \\ \Sigma_{YW} & \Sigma_{YY} \end{bmatrix} \right)$$

We find the covariances:

- $\Sigma_{YY} = \mathbb{E}[YY^T] = \mathbb{E}[(XW + \sqrt{\lambda}N)(XW + \sqrt{\lambda}N)^T] = XIX^T + \lambda I = XX^T + \lambda I$
- $\Sigma_{WY} = \mathbb{E}[WY^T] = \mathbb{E}[W(XW + \sqrt{\lambda}N)^T] = \mathbb{E}[WW^T]X^T + 0 = IX^T = X^T$

From the formula for the conditional expectation of jointly Gaussian random variables,

$$\mathbb{E}[W | Y = y] = \Sigma_{WY} \Sigma_{YY}^{-1} y = X^T (XX^T + \lambda I)^{-1} y$$

Thus, the MAP estimate for W given $Y = y$ is:

$$\hat{w} = X^T (XX^T + \lambda I)^{-1} y \quad (6)$$

We can show that this is algebraically equivalent to:

$$\hat{w} = (X^T X + \lambda I)^{-1} X^T y \quad (5)$$

using the Matrix Inversion Lemma

$$X^T (XX^T + \lambda I)^{-1} = (X^T X + \lambda I)^{-1} X^T$$

Hence, both expressions are equivalent, and the MAP interpretation yields the same solution as minimizing the regularized least squares objective.

Problem 4

(a)

We are given the optimization problem:

$$\min_{\mathbf{w}} \mathbb{E} \left[\|\tilde{X}\mathbf{w} - \mathbf{y}\|^2 \right]$$

Let $\tilde{X}_i = x_i + N_i$ where $N_i \sim \mathcal{N}(0, \sigma^2 I_n)$ are i.i.d. noise vectors.

Expanding the expectation using linearity:

$$\mathbb{E} \left[\|\tilde{X}\mathbf{w} - \mathbf{y}\|^2 \right] = \sum_{i=1}^m \mathbb{E} \left[(\tilde{x}_i^\top \mathbf{w} - y_i)^2 \right]$$

Substitute $\tilde{x}_i = x_i + N_i$:

$$\mathbb{E} \left[((x_i + N_i)^\top \mathbf{w} - y_i)^2 \right] = \mathbb{E} \left[(x_i^\top \mathbf{w} - y_i + N_i^\top \mathbf{w})^2 \right]$$

Using $\mathbb{E}[N_i^\top \mathbf{w}] = 0$ and $\mathbb{E}[(N_i^\top \mathbf{w})^2] = \sigma^2 \|\mathbf{w}\|^2$, we get:

$$\mathbb{E} \left[(\tilde{x}_i^\top \mathbf{w} - y_i)^2 \right] = (x_i^\top \mathbf{w} - y_i)^2 + \sigma^2 \|\mathbf{w}\|^2$$

Summing over all i :

$$\mathbb{E} \left[\|\tilde{X}\mathbf{w} - \mathbf{y}\|^2 \right] = \|X\mathbf{w} - \mathbf{y}\|^2 + m\sigma^2 \|\mathbf{w}\|^2$$

Therefore, minimizing this is equivalent to:

$$\min_{\mathbf{w}} \frac{1}{m} \|X\mathbf{w} - \mathbf{y}\|^2 + \lambda \|\mathbf{w}\|^2 \quad \text{where } \lambda = \sigma^2$$

(b)

Let $x \in \mathbb{R}$ be the scalar input and $y \in \mathbb{R}$ be the label. At iteration t , the noisy input is $\tilde{x} = x + N_t$, with $N_t \sim \mathcal{N}(0, \sigma^2)$. The squared loss is:

$$L(w_t) = \frac{1}{2} (\tilde{x}w_t - y)^2$$

Gradient:

$$\frac{\partial L}{\partial w_t} = (\tilde{x}w_t - y)\tilde{x}$$

Gradient descent update:

$$w_{t+1} = w_t - \eta(\tilde{x}w_t - y)\tilde{x}$$

Taking expectation over the noise:

$$\mathbb{E}[w_{t+1}] = \mathbb{E}[w_t - \eta(\tilde{x}w_t - y)\tilde{x}] = \mathbb{E}[w_t - \eta((x + N_t)w_t - y)(x + N_t)]$$

Expanding and using $\mathbb{E}[N_t] = 0$, $\mathbb{E}[N_t^2] = \sigma^2$:

$$\mathbb{E}[w_{t+1}] = w_t - \eta \left[(x^2 + \sigma^2)w_t - xy \right]$$

So:

$$\mathbb{E}[w_{t+1}] = (1 - \eta(x^2 + \sigma^2)) \mathbb{E}[w_t] + \eta xy$$

(c)

For convergence of $\mathbb{E}[w_t]$, we need the coefficient of $\mathbb{E}[w_t]$ to be in $(-1, 1)$:

$$|1 - \eta(x^2 + \sigma^2)| < 1 \Rightarrow 0 < \eta < \frac{2}{x^2 + \sigma^2}$$

(d)

Let $a = 1 - \eta(x^2 + \sigma^2)$ and $b = \eta xy$. This recurrence:

$$\mathbb{E}[w_{t+1}] = a\mathbb{E}[w_t] + b$$

is a first-order linear difference equation with fixed point solution of

$$\lim_{t \rightarrow \infty} \mathbb{E}[w_t] = \frac{b}{1 - a} = \frac{\eta xy}{1 - (1 - \eta(x^2 + \sigma^2))} = \frac{xy}{x^2 + \sigma^2}$$

If there were no noise ($\sigma^2 = 0$), then the recurrence becomes:

$$\mathbb{E}[w_{t+1}] = (1 - \eta x^2)\mathbb{E}[w_t] + \eta xy$$

and the limiting value becomes:

$$\lim_{t \rightarrow \infty} \mathbb{E}[w_t] = \frac{xy}{x^2} = \frac{y}{x}$$

Adding noise to the input data results in a limiting value of the learned weight:

$$\boxed{\lim_{t \rightarrow \infty} \mathbb{E}[w_t] = \frac{xy}{x^2 + \sigma^2}}$$

This is strictly smaller in magnitude than the noiseless solution when $\sigma^2 > 0$. This indicates that the learned weight is effectively shrunk toward zero, which is characteristic of ℓ_2 regularization (ridge regression).